

CE 557: Solid Mechanics in structural engineering

Course Code	CE 557
Course Name	Solid Mechanics in structural engineering
L-T-P-C	3-0-0-3
Prerequisites	Strength of Material and Structures (CE 301) or Mechanics of Solids (ME206) or Equivalent
Intended for	UG (Civil)/ M.S./ M.Tech./ Ph.D.
Distribution	Specialization Core (M.Tech. in Structural Engineering)/Elective (UG, MS, PhD, M.Tech)
Semester	Odd/Even
Approval	23rd BoA (Apr 2018); 16th Senate (Apr 2018)

Course Contents

- **Theories of Stress and Strain:** Stress at a point, Equilibrium equations, Deformation of a Deformable Body, Strain tensor, Transformation of Stress/Strain, Principal Stress/Strains, Strain Compatibility, Strain ' Displacement relations in Cylindrical coordinates, Equilibrium equations in Cylindrical Coordinates, Strain measurement and Strain Rosettes, Hooke's Law: Isotropic Elasticity, Elastic constants and their relations, Displacement Equations of Equilibrium. [8 Lectures]
- **Plane Strain and Plane Stress:** Plane Stress, Plane Strain, Airy's Stress function, Differential equation for the stress function, Polar coordinates in the plane, Bending of Cantilever beam, Thick ' walled Cylinder subjected to Internal and External Pressures - Lamé's Problem. [5 Lectures]
- **Bending of Beams:** Nonsymmetrical bending, deflections of beams subjected to Nonsymmetrical bending, Shear flow in thin Wall beam cross section, Bending of Curved Beams (Winkler ' Bach Formula). [7 Lectures]
- **Torsion:** Torsion of Prismatic Bars, Torsion of Circular, Elliptical, Equilateral Triangular and Rectangular bars, Membrane Analogy, Torsion of Thin ' walled tubes. [6 Lectures]
- **Beams on Elastic Foundations:** Infinite beam subjected to a Concentrated/ Distributed Load, Semi - infinite Beam subjected to end loads, short beams. [5 Lectures]
- **Failure Theories and Introduction to Ideally Plastic solid:** (6 hours) Theories of failure, Factor of Safety in Design, Mohr's Theory of failure, Ideally Plastic solid, Yield Surfaces of Tresca and Von Mises, Stress-Strain Relations (Plastic Flow), Prandtl- Reuss Equation. [6 Lectures]
- **Three-Dimensional elastic half-space:** Elastic half-space subjected to surface point, and distributed loads (Boussinesq problem), Propagation of waves in elastic half-space (dilatation, distortion and surface waves), Introduction to earthquake induced ground vibration. [5 Lectures]

Text Books

1. Srinath L.S, **Advanced Mechanics of Solids**, 3rd Edition, Tata McGraw-Hill, 2008.
2. Timoshenko, S.P. and Goodier, J.N, **Theory of Elasticity**, 3rd Edition, McGraw- Hill, 2010.

References

1. K. F. Graff, **Wave Motion in Elastic Solids**, Dover Publications, 1991.
2. M. H. Sadd, **Elasticity: Theory, application and numerics**, 3rd Edition, Elsevier India, 2014.